Machine Learning Prediction of Heating and Cooling Loads of Residential Buildings

Project Plan

Oğuz Kağan Göçmez

***Abstract*—Heating and cooling loads determine the size of the heating, ventilation and air conditioning (HVAC) equipment that a building needs. They are usually computed with detailed simulation software, which is slow and requires expert users. Machine learning (ML) models trained on simulated buildings can give the same estimates in a fraction of a second. In this project we use the UCI Energy Efficiency dataset, which contains 768 residential buildings simulated in Ecotect and described by eight design parameters (relative compactness, surface area, wall area, roof area, overall height, orientation, glazing area and glazing area distribution). We predict two outputs: the heating load (HL) and the cooling load (CL). Following the reference study of Tsanas and Xifara, we will explore the data statistically, compare six regression models with repeated 10-fold cross-validation and three error measures, tune the most promising models, and analyze which inputs matter most.**

As our own contribution, we will test how well the models generalize to building shapes that were not seen during training. In a random split, buildings that differ only in orientation, which mostly have very similar loads, end up in both the training and the test sets, which may make the reported errors optimistic. Our first experiments already reproduce the statistical analysis of the reference study, and an untuned random forest reaches a mean absolute error of 0.31 for HL and 0.98 for CL, compared with 0.51 and 1.42 in the reference study.

***Index Terms*—Building energy performance, heating load, cooling load, machine learning, regression, cross-validation.**

I. Introduction

HVAC systems account for most of the energy used in buildings, so estimating heating and cooling loads early in the design phase helps to design more energy-efficient buildings [1]. Simulation tools give reliable estimates but are time-consuming, which has motivated the use of ML models such as decision trees [2], support vector machines [3] and polynomial regression [4].

Tsanas and Xifara [1] published the dataset used in this project [5]. They compared a robust linear model, iteratively reweighted least squares (IRLS), with a random forest (RF), and reported mean absolute errors (MAE) of 0.51 for HL and 1.42 for CL with the RF. Later studies applied other artificial intelligence techniques [6] and genetic programming [7] to the same data. The dataset is also a common benchmark: Žegklitz and Pošík [8] used it to compare symbolic regression methods with a tuned RF and support vector regression (SVR), and Yazici *et al.* [9] used it to test ML models on a low-power Internet of Things device. We follow the Machine Learning Project Checklist of the course textbook [10].

The author is with Özyeğin University, İstanbul, Türkiye. This plan is part of the EE 311 Introduction to Artificial Intelligence, Machine Learning and Data Science course project, Fall 2026.

TABLE I
Project Schedule

Subtask	Week	Status
Literature search	1–2	In progress
Frame the problem	1	Done
Get the data	1	Done
Explore the data	2	Done
Project plan report	2	This report
Prepare the data	3	Planned
Shortlist models (random CV)	3–4	First pass
Grouped CV by building shape	5	Planned
Midterm report	6	Planned
Fine-tune the best models	7–8	Planned
Feature importance and errors	9	Planned
Final tables and figures	10	Planned
Final report	11–12	Planned
Recorded presentation	12	Planned

Week 1 is the week the topic was selected; week 12 is the last week of the semester. All subtasks are carried out by the author.

II. Project Schedule

Table I lists the subtasks with their deadlines. This is a single-person project, so the author is responsible for all subtasks. The approach for each subtask is described below.

A. Literature Search

We read the reference study [1] and its references, and we search for later work on the same dataset (for example, papers citing [1] on Google Scholar) to collect results to compare against. This continues until the final report.

B. Frame the Problem

The task is supervised regression with two continuous outputs, HL and CL. To make our results directly comparable with [1], we use the same three errors: mean absolute error (MAE), mean squared error (MSE) and mean relative error (MRE). The IRLS and RF results of [1] are our baseline.

C. Get the Data

We load the dataset [5] from a CSV export of the original Excel file, which uses semicolons as separators and commas as decimal points. We checked it against the Excel file: 768 buildings, 10 columns, no missing values and no duplicate rows.

TABLE II
FIRST RESULTS: MAE (MEAN ± STD) WITH 10-FOLD CV REPEATED 10 TIMES, UNTUNED MODELS

Model	Heating load	Cooling load
Linear regression	2.09 ± 0.24	2.27 ± 0.27
Robust linear (Huber)	2.03 ± 0.25	2.20 ± 0.30
k-nearest neighbors	1.42 ± 0.20	1.66 ± 0.24
Support vector regression	0.55 ± 0.08	1.13 ± 0.19
Gradient boosting	0.34 ± 0.04	1.02 ± 0.13
Random forest	$\mathbf{0.31 \pm 0.04}$	$\mathbf{0.98 \pm 0.14}$
IRLS in [1]	2.14 ± 0.24	2.21 ± 0.28
RF in [1]	0.51 ± 0.11	1.42 ± 0.25

D. Explore the Data

We plot histograms and input–output scatter plots, and we measure the association of each input with each output using the Spearman rank correlation (with p-values) and mutual information, as in [1]. This step is done, and our Spearman correlations match those of [1]. Relative compactness and surface area are perfectly rank-correlated ($\rho = -1$), so they carry the same information. Orientation is not correlated with either output, although it does change the cooling load of some two-storey shapes, in a way that does not go in one direction.

E. Prepare the Data

No cleaning is needed. Inputs are standardized inside a pipeline, so the scaler is fitted on the training folds only. We will also test one-hot encoding of orientation and glazing area distribution, and removing the redundant surface area input.

F. Shortlist Promising Models

We compare six models implemented in scikit-learn [11]: linear regression, robust linear regression (Huber loss, our counterpart to IRLS), k-nearest neighbors, support vector regression, random forest [12] and gradient boosting [13]. They are evaluated with 10-fold cross-validation (CV) repeated on random splits, as in [1]. As our contribution, we repeat the evaluation with grouped CV [14], where all buildings of the same shape (one of 12) stay in the same fold, to measure performance on unseen shapes. Table II shows the first, untuned results.

G. Fine-Tune the System

For the two or three best models we will search the hyperparameters (grid or randomized search inside CV). The final results will use 100 CV repetitions, as in [1]. We will also compute permutation importance and add a random probe feature, as in [1], to check which inputs matter most, and we will try a single model that predicts both outputs together.

H. Present the Results

We will summarize the results in tables and figures, compare them with [1], [6], [7] and [8], and prepare the midterm report, the final report and a recorded presentation of 10–15 minutes.

ACKNOWLEDGMENT

AI tools (Claude, Anthropic) were used to help draft this plan and the initial analysis code. All suggestions were reviewed carefully by the author, and their correctness was validated.

REFERENCES

[1] A. Tsanas and A. Xifara, "Accurate quantitative estimation of energy performance of residential buildings using statistical machine learning tools," *Energy Build.*, vol. 49, pp. 560–567, 2012, doi: 10.1016/j.enbuild.2012.03.003.

[2] Z. Yu, F. Haghighat, B. C. M. Fung, and H. Yoshino, "A decision tree method for building energy demand modeling," *Energy Build.*, vol. 42, no. 10, pp. 1637–1646, 2010, doi: 10.1016/j.enbuild.2010.04.006.

[3] B. Dong, C. Cao, and S. E. Lee, "Applying support vector machines to predict building energy consumption in tropical region," *Energy Build.*, vol. 37, no. 5, pp. 545–553, 2005, doi: 10.1016/j.enbuild.2004.09.009.

[4] T. Catalina, J. Virgone, and E. Blanco, "Development and validation of regression models to predict monthly heating demand for residential buildings," *Energy Build.*, vol. 40, no. 10, pp. 1825–1832, 2008, doi: 10.1016/j.enbuild.2008.04.001.

[5] A. Tsanas and A. Xifara, "Energy Efficiency," UCI Machine Learning Repository, 2012. [Online]. Available: https://doi.org/10.24432/C51307

[6] J.-S. Chou and D.-K. Bui, "Modeling heating and cooling loads by artificial intelligence for energy-efficient building design," *Energy Build.*, vol. 82, pp. 437–446, 2014, doi: 10.1016/j.enbuild.2014.07.036.

[7] M. Castelli, L. Trujillo, L. Vanneschi, and A. Popovič, "Prediction of energy performance of residential buildings: A genetic programming approach," *Energy Build.*, vol. 102, pp. 67–74, 2015, doi: 10.1016/j.enbuild.2015.05.013.

[8] J. Žegklitz and P. Pošík, "Symbolic regression algorithms with built-in linear regression," 2017, arXiv:1701.03641. [Online]. Available: https://arxiv.org/abs/1701.03641

[9] M. T. Yazici, S. Basurra, and M. M. Gaber, "Edge machine learning: Enabling smart Internet of Things applications," *Big Data Cogn. Comput.*, vol. 2, no. 3, Art. no. 26, 2018, doi: 10.3390/bdcc2030026.

[10] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022.

[11] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: https://jmlr.org/papers/v12/pedregosa11a.html

[12] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[13] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, 2001, doi: 10.1214/aos/1013203451.

[14] D. R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017, doi: 10.1111/ecog.02881.